

NBA Player Performance Analysis

Jobally Forde

Introduction & Problem Statement

- The goal of this project is to analyze the factors influencing NBA player performance during the 2023-2024 regular season and playoffs. The analysis will identify key performance metrics that significantly impact player effectiveness. Insights derived from this project will provide actionable recommendations to inform strategic decisions, player training, and game strategies.

Context

- In the highly competitive environment of the NBA, understanding what drives player performance is crucial for teams aiming to optimize their strategies, improve player training, and enhance overall team effectiveness. Insights from the data can provide teams and analysts with information to help improve player effectiveness during both the regular season and playoffs.

Criteria for Success

- The project will be considered successful if it uncovers key performance metrics that offer actionable and valuable insights for NBA teams, analysts, and coaches. These insights should help optimize player performance and improve strategic planning during games.

Scope of Solution Space

- The analysis focuses on player data from the 2023-2024 season, covering both the regular season and playoffs. This project will use statistical analysis and data visualization techniques to uncover patterns, trends, and performance drivers. The aim is to isolate individual player contributions, despite team dynamics, using the available data.

Constraints

- Key constraints include potential issues with data quality, challenges in isolating individual contributions from team dynamics, and the necessity of preprocessing the data.

Stakeholders

- The primary stakeholders are NBA teams, coaches, and analysts who will use the findings to make strategic decisions, optimize training programs, and improve team performance.

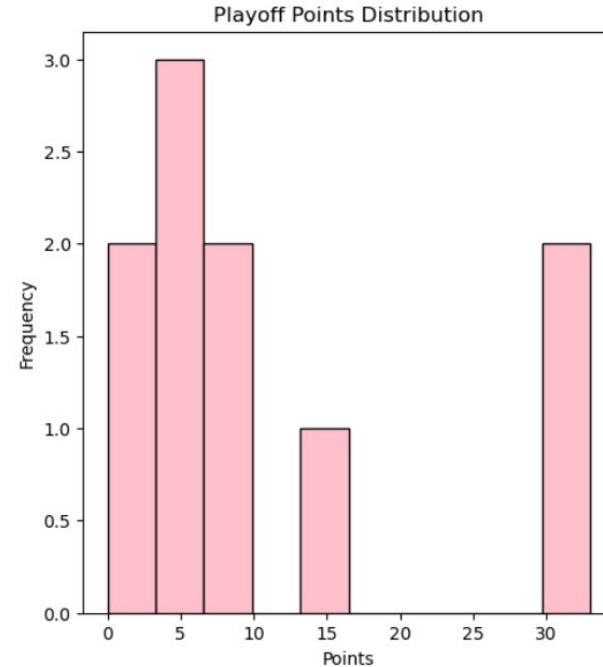
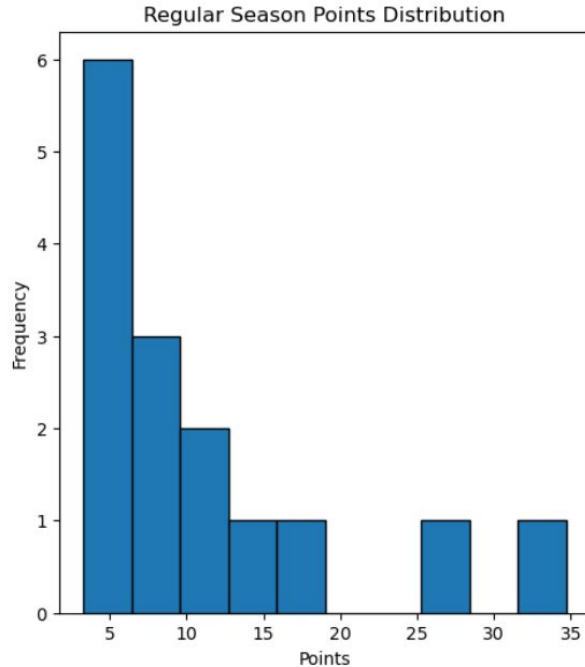
Data Sources

- The dataset used for this analysis is *NBA Player Stats Season 23/24* sourced from Kaggle. The dataset contains detailed statistics of player performance across both regular season and playoff games, including metrics such as points, assists, rebounds, and shooting percentages.

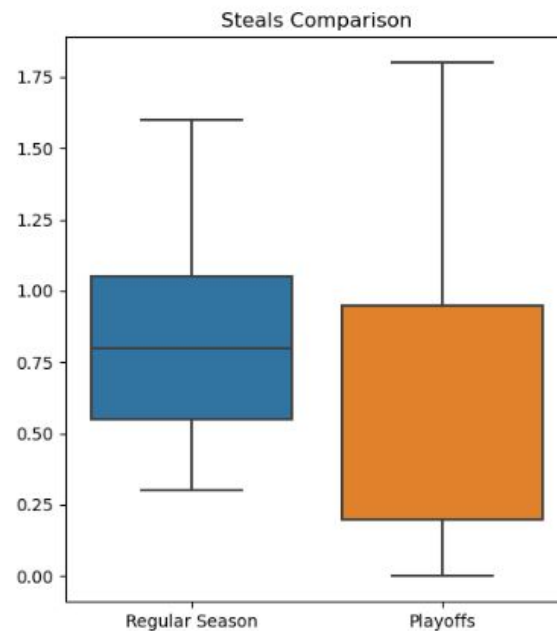
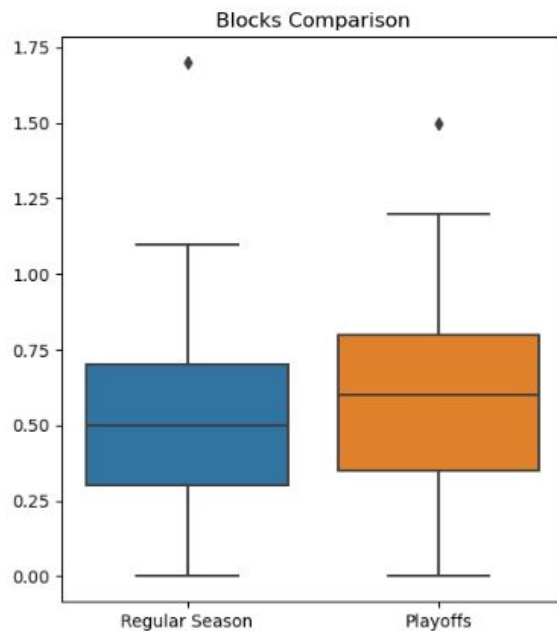
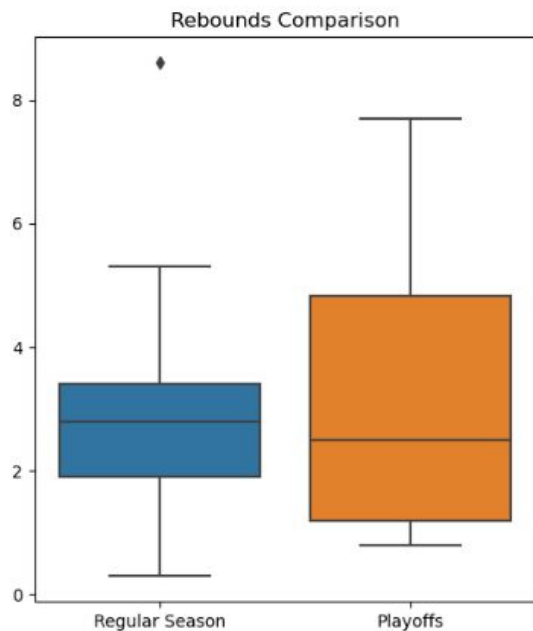
Exploratory Data Analysis (EDA)

- The performance of the 76ers players in the regular season shows no significant statistical differences in defensive attributes like rebounds, blocks and steals compared to the playoffs.
- Key players like Joel Embiid and Tyrese Maxey maintained or increased their contributions, while Tobias Harris showed a decline in points in playoffs performance.
- Overall there wasn't any significant difference between 76ers performance in the regular season and playoffs.

Points Scored by team in the Regular season vs Playoffs



Defensive Attribute Comparisons



Modeling Overview

The primary objective of the modeling phase was to predict NBA player performance based on key metrics from both the regular season and the playoffs in the 2023-2024 NBA season. Three models were employed to approach this problem:

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor

Linear Regression

Linear Regression was used as a baseline model to understand the relationship between player performance metrics and player effectiveness.

Regular Season:

MSE: 1.4016

R-squared: 0.9818

Playoffs:

MSE: 9.4382

R-squared: 0.9497

The Linear Regression model performed very well on the regular season data, capturing over 98% of the variance. The performance in the playoffs was slightly lower, but still strong, with an R-squared value of 0.95.

Data Splitting

The data was split into training and test sets for both regular season and playoff games:

- Regular Season: The dataset was split with an 80-20 train-test ratio.
- Playoffs: The playoff dataset followed the same 80-20 split pattern.

Linear Regression

Linear Regression was used as a baseline model to understand the relationship between player performance metrics and effectiveness.

Results:

- Regular Season:
- MSE: 1.4016
- R-squared: 0.9818

- Playoffs:
- MSE: 9.4382
- R-squared: 0.9497

Linear Regression performed exceptionally well on the regular season data, capturing over 98% of the variance. While performance in the playoffs was slightly lower, it still achieved a strong R^2 value of 0.95.

Random Forest

The Random Forest model, an ensemble-based method, was employed to handle complex relationships between variables.

Results:

- Regular Season:
- MSE: 39.1977
- R-squared: 0.4913

- Playoffs:
- MSE: 107.4397
- R-squared: 0.4276

Despite being a powerful model, Random Forest underperformed compared to Linear Regression, with lower R-squared values and higher MSE for both the regular season and playoffs.

Gradient Boosting

Gradient Boosting was explored as an alternative model, known for handling small datasets and noisy data more effectively.

Results:

- Regular Season:
- MSE: 28.0834
- R-squared: 0.6355

- Playoffs:
- MSE: 14.9114
- R-squared: 0.9206

Gradient Boosting outperformed Random Forest, especially for the playoff data, achieving an R^2 score of 0.92. However, it still lagged behind Linear Regression in terms of regular season performance.

Model Comparison

- In the model comparison, Linear Regression outperforms others in the regular season with an MSE of 1.4016 and an R-squared of 0.9818, while also showing strong playoff performance (MSE of 9.4382, R-squared of 0.9497). Random Forest has poor performance (MSE: 39.1977, R-squared: 0.4913 in the regular season; MSE: 107.4397, R-squared: 0.4276 in the playoffs), and Gradient Boosting shows moderate results (MSE: 28.0834, R-squared: 0.6355 in the regular season; MSE: 14.9114, R-squared: 0.9206 in the playoffs).
- Linear Regression emerged as the best-performing model for both the regular season and playoffs based on R-squared and MSE values. Gradient Boosting was the second-best model, particularly for the playoff data. Random Forest underperformed, suggesting that simpler models like Linear Regression may be more effective when the dataset has a clear linear relationship.

Conclusion and Recommendations

- Key Performance Drivers: Based on the analysis, metrics like points, assists, and shooting percentages were found to be strong predictors of player performance.
- Modeling Insights: Linear Regression, the simplest model, performed the best, indicating that player performance during the regular season and playoffs can be explained by relatively straightforward relationships between metrics.
- Strategic Recommendations: NBA teams and analysts can use these insights to focus on training and improving key performance areas like shooting accuracy and assist-making, which have a significant impact on player effectiveness.
- Future Research: Further studies could explore additional factors such as player fatigue, game conditions, and psychological aspects that might affect performance.